Q41 (W25 Final)

IV estimator: $\text{plim}\ \hat{\beta}^{IV} = \beta + \dfrac{Cov(z,u)}{Cov(z,x)}$.

Want to show $\hat{\beta}^{IV}$ is consistent w/. a valid inst.

Consistency: (1) The est. has a prob. limit
(2) The plim = the parameter of interest

$\hat{\beta}^{IV}$ being consistent $\Rightarrow$ $\text{plim}\ \hat{\beta}^{IV} = \beta$

exog. of inst.

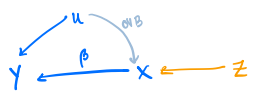
Need + $\boxed{Cov(z,u) = 0}$
+ $Cov(z,x) \neq 0$

$\Rightarrow \text{plim}\ \hat{\beta}^{IV} = \beta + \dfrac{0}{\neq 0} = \beta$

relevance of inst.

$y = \beta_0 + \beta_1 x + u$

worried about OVB... could find an instrumental var z



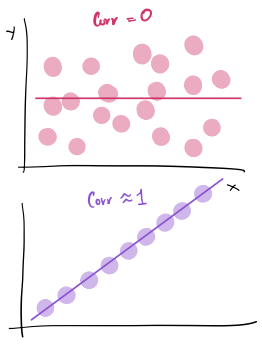
Cov. vs Corr.

$Corr(Y,X) = \dfrac{Cov(Y,X)}{\sigma_Y \cdot \sigma_X}$

$-\infty, \infty$

unitless & $[-1, 1]$

≥ 0 (+) $\quad \geq 0$



Panel data and FEs

Cross-sectional data: ind. i from 1 to n in a given time period

$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$

$Inc_i = \beta_0 + \beta_1 Educ_i + \beta_2 Health_i + \beta_3 ParentalWealth_i + u_i$

Time-series data: one ind./unit for time periods $t = 1, \ldots, T$

$y_t = \beta_0 + \beta_1 x_t + u_t$ ← Static model

$Unemp_t = \beta_0 + \beta_1 Policy_t + \cdots + u_t$

$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + u_t$ dynamic model w/. lagged expl. vars

$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \beta_3 y_{t-1} + u_t$ dynamic model w/. lagged outcome and lagged expl. ADL(1,1)

one more concern: Autocorrelation:

$u_t = \rho u_{t-1} + \varepsilon_t$

$-1 < \rho < 1$

Now, panels

↳ combine cross-sections <u>AND</u> time-series

i	t
1	1
2	1
3	1
4	1
⋮	⋮
n	1
1	2
2	2
3	2
⋮	⋮
n	n
⋮	⋮
1	T
2	T
3	T
⋮	⋮
n	n

A cross section $(t=1)$

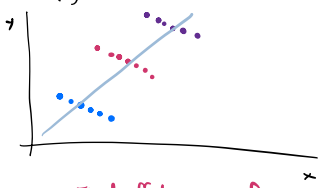
A cross-section $(t=2)$

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it}$$

OLS on β_1 here reflects:

(A) within-ind. changes through time,

(B) cross-ind. diffs in Y (and X)



Solution: Fixed effects account for

(1) diffs btn individuals (static in time),

(2) shared shocks in time.

Plain OLS: $y_{it} = \beta_0 + \beta_1 x_{it} + u_{it}$

Fixed effects model: $y_{it} = \alpha_i + \gamma_t + \beta_1 x_{it} + u_{it}$

"Fixed effect" for each ind. i (an intercept for each ind)

"Fixed effect" for time period t (an intercept for each period in time)

Panels also <u>can</u> help w/. non-stationarity...

(Non)-stationarity

Stationarity is a way of saying our data behave well through time.

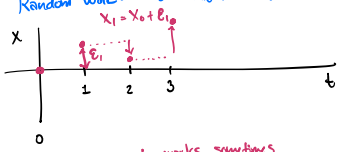
We want variable that are

(1) mean stationary: the mean is stable through time, ie, $E[X_t] = E[X_{t+k}]$

(2) variance stationary: the variance is stable through time, ie, $V[X_t] = V(X_{t+k})$

(3) covariance stationary: $Cov(X_t, X_{t+k}) = Cov(X_s, X_{s+k})$

Random walk: $X_t = X_{t-1} + \varepsilon_t$



$$Var(X_1) = Var(\underbrace{X_0}_{\text{constant}} + \varepsilon_1) = \sigma^2$$

$Var(\varepsilon_1) = \sigma^2$

$$Var(X_2) = Var(X_1 + \varepsilon_2) = \sigma^2 + \sigma^2 = 2\sigma^2$$

$\vdots$

$$Var(X_t) = t \cdot \sigma^2$$

violates variance stationarity

not stable in time

(it increases in time)

only works sometimes

Solution: differencing

$\Delta y_t = \beta_1 \Delta x_t + \Delta u_t$ does not suffer from non-stationarity if y & x are difference-stationary (eg, random walks)